

Main Takeaways

the new version fix most of things (problem of different extraction date in previous version causing duplicate/spikes and mapping between one item to multiple categories)

1. the difference between soda away and at home closes but still different from Leo (8 times larger instead of 5-6 for 2023);
 - Leo's level (2,859,563,677 versus 477,718,593.9), here (4,115,420,727.33 versus 513,453,321.20);
 - soda away is similar but soda at home is 1.4 time larger than Leo's number;
 - either there are other changes/caveats that I did not notice or the weight is different from Leo's (as Heather from Kilts says trend factor could be wrong);
 - And national factor is constant between household, so the closure of gap could be mostly due to trend factor;
2. Increasing trend in expenditure; seasonal effect around end of year (increase in food spending and relative proportion of total spending on food); same for soda spending;
 - Could be due to inflation
 - this seasonal effect (around December) is not shown in Nielsen's trend analysis -> limited capture of food away from home?
 - No significant increasing trend in Nielsen (measured in volume instead of monetary unit)

Figure 1: Monthly Average of Total Spending

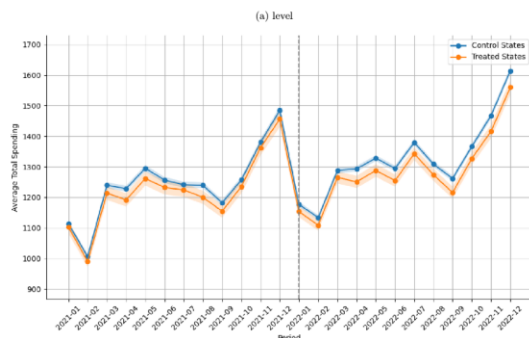
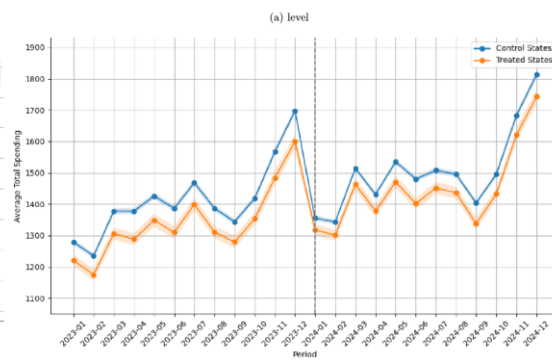


Figure 1: Monthly Average of Total Spending



3. Trend in EBT spending -> policy changes in SNAP benefits (see Numerator's estimates of SNAP benefit issued by month; SNAP benefit increased in Sep 2022 and decreased in Jan 2023)

Figure 3: Monthly Average of Food Spending on EBT

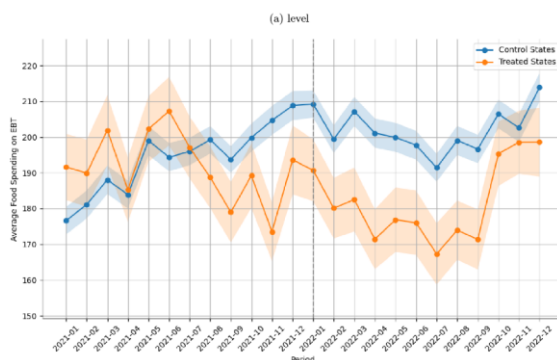
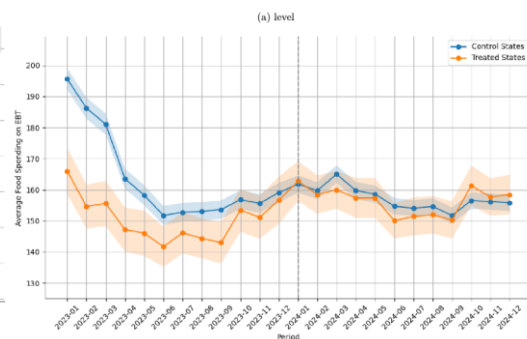
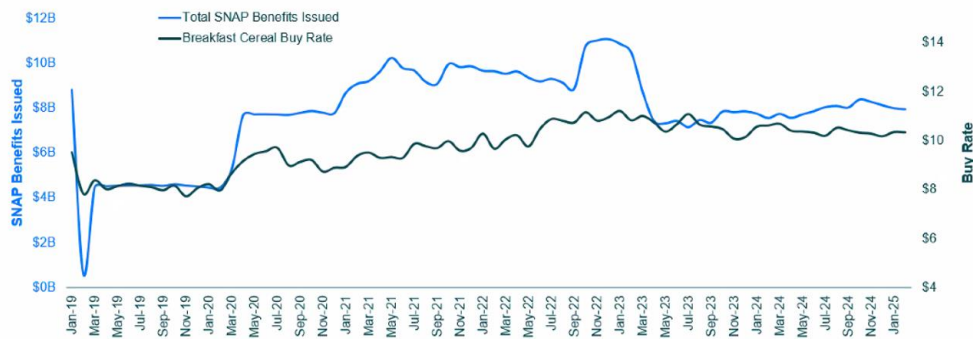


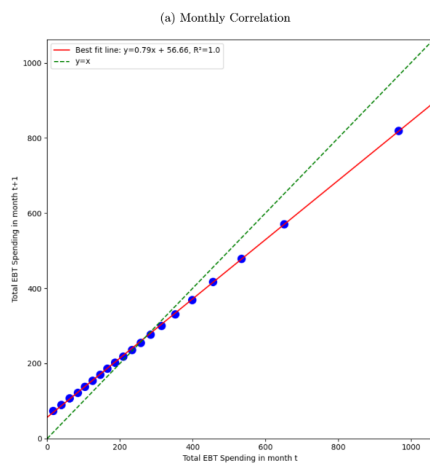
Figure 3: Monthly Average of Food Spending on EBT



Total SNAP Benefits Issued by Month vs Breakfast Cereal Buy Rate



4. bin scatter -> much closer to 1 after conditional on those with consecutive SNAP spell ≥ 12



5. working on consumption by household attributes (diabetes): people with diabetes spend (report) more on food and soda and have higher EBT spending, but they have a lower share of soda paid by EBT than those without diabetes (are less likely to pay soda by EBT cash than those without diabetes); 18% if SNAP household having diabetes, 16% of all having diabetes

	Mean (SD)					Min	Max
	SNAP		Non-SNAP		Total		
	Diabetes	No Diabetes	Diabetes	No Diabetes			
Food Spending	718.31 (514.87)	675.50 (507.37)	619.19 (424.46)	585.52 (426.99)	610.31 (442.36)	0.00	50429.40
Total Spending	1595.87 (1161.51)	1507.97 (1153.81)	1565.93 (1110.22)	1485.95 (1128.68)	1513.52 (1129.46)	0.00	135684.55

	Mean (SD)					Min	Max
	SNAP		Non-SNAP		Total		
	Diabetes	No Diabetes	Diabetes	No Diabetes			
Soda Spending	34.25 (43.39)	29.43 (39.30)	26.61 (39.34)	20.53 (32.63)	23.67 (35.96)	0.00	5749.22
Prop of Soda Spending on EBT	0.30 (0.34)	0.33 (0.36)	0.15 (0.27)	0.16 (0.29)	0.28 (0.35)	0.00	1.00

Figure 11: Proportion of Food Spending on EBT

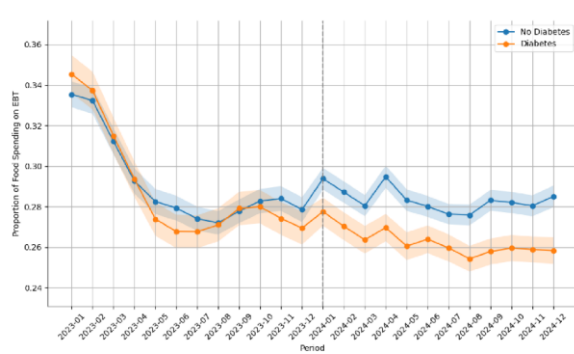
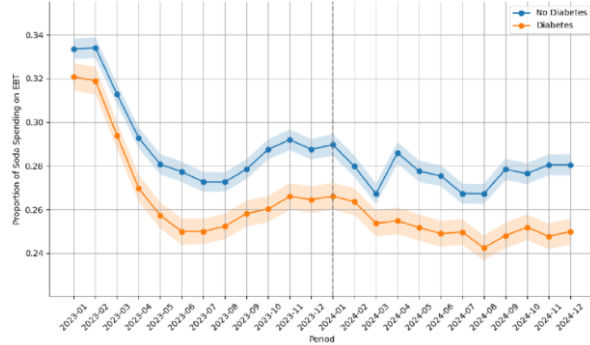


Figure 14: Proportion of Soda Spending on EBT



6. compare to CEX

- Annual total expenditure is severely underestimated but the difference between food spending are much smaller -> Numerator capture more on grocery and food expenditure than other type of expenditure; richer household's expenditure is more severely underestimated than others

	Total	<15k	15-29k	30-39k	40-49k	50-69k	70-99k	100-149k	150-199k	200k+
Total Spending	16,956.61 (22,970.10)	14,921.84 (21,658.75)	14,130.09 (19,978.39)	13,907.81 (18,461.29)	14,160.82 (18,606.28)	14,439.09 (19,847.39)	15,480.91 (20,111.03)	18,734.63 (26,562.45)	24,901.28 (27,652.42)	25,960.63 (29,257.26)
Food Spending	6,758.58 (9,108.25)	6,297.14 (9,181.03)	6,007.39 (9,336.73)	5,764.50 (7,939.10)	5,830.90 (7,650.04)	5,859.87 (9,246.84)	6,123.58 (8,033.65)	7,255.13 (9,333.09)	9,533.27 (10,102.21)	9,906.80 (10,942.92)

Item	All consumer units	Less than \$15,000	\$15,000 to \$29,999	\$30,000 to \$39,999	\$40,000 to \$49,999	\$50,000 to \$69,999	\$70,000 to \$99,999	\$100,000 to \$149,999	\$150,000 to \$199,999	\$200,000 and more
Average annual expenditures										
Mean	\$77,280	\$32,081	\$35,587	\$47,027	\$50,204	\$59,461	\$71,899	\$90,677	\$117,818	\$174,993
SE	994.16	1,385.24	1,054.11	1,212.88	1,058.24	1,163.64	1,243.75	1,218.89	2,832.90	3,968.96
RSE	1.29	4.32	2.96	2.58	2.11	1.96	1.73	1.34	2.40	2.27
Food										
Mean	9,985	5,684	5,181	6,665	7,302	8,413	9,705	11,960	15,264	18,525
Share	12.9	17.7	14.6	14.2	14.5	14.1	13.5	13.2	13.0	10.6
SE	122.90	328.22	233.61	323.48	370.02	222.37	308.20	207.67	464.66	493.89
RSE	1.23	5.77	4.51	4.85	5.07	2.64	3.18	1.74	3.04	2.67

- In Numerator, share of food is more constant over income than in CEX (could be result of financial incentives distributed by Numerator; poorer household have more incentive to scan receipts; expenditure of richer household is not well captured)
- FAH more precise than FAFH -> in general QSR receipts are assigned with less financial reward; full restaurant services not well captured according to Numerator

Food at Home	6,144.95 (8,486.20)	5,674.39 (8,486.89)	5,428.77 (8,824.30)	5,193.74 (7,346.39)	5,250.15 (7,003.62)	5,291.89 (8,698.08)	5,556.99 (7,397.79)	6,637.01 (8,736.01)	8,756.87 (9,443.60)	9,132.91 (10,189.02)
Food Away from Home	613.63 (1,207.37)	622.76 (1,244.22)	578.63 (1,173.66)	570.76 (1,084.51)	580.75 (1,166.10)	567.98 (1,162.19)	566.59 (1,125.24)	618.12 (1,153.94)	776.41 (1,318.74)	773.89 (1,618.36)
Soda Spending	243.67 (609.29)	263.68 (587.44)	258.81 (512.47)	239.28 (448.09)	226.73 (447.16)	223.95 (1,033.27)	218.92 (439.49)	242.78 (519.88)	295.68 (491.08)	287.38 (545.79)

Food at home										
Mean	6,053	4,037	3,624	4,430	4,769	5,536	6,080	7,053	8,622	9,702
Share	7.8	12.6	10.2	9.4	9.5	9.3	8.5	7.8	7.3	5.5
SE	84.91	199.09	185.00	286.97	258.02	181.16	229.41	168.09	258.50	291.30
RSE	1.40	4.93	5.10	6.48	5.41	3.27	3.77	2.38	3.00	3.00

Item	All consumer units	Less than \$15,000	\$15,000 to \$29,999	\$30,000 to \$39,999	\$40,000 to \$49,999	\$50,000 to \$69,999	\$70,000 to \$99,999	\$100,000 to \$149,999	\$150,000 to \$199,999	\$200,000 and more
Nonalcoholic beverages										
Mean	663	490	434	514	572	620	677	770	856	976
Share	.9	1.5	1.2	1.1	1.1	1.0	.9	.8	.7	.6
SE	12.38	36.54	33.79	38.74	43.37	24.11	35.06	30.93	39.55	49.02
RSE	1.87	7.45	7.78	7.53	7.59	3.89	5.18	4.02	4.62	5.02
Food prepared by consumer unit on out of town trips										
Mean	96	31	31	53	57	64	72	112	162	265
Share	.1	.1	.1	.1	.1	.1	.1	.1	.1	.2
SE	4.96	4.85	6.80	11.24	12.77	6.17	7.15	10.78	15.04	22.98
RSE	5.17	15.64	21.63	21.05	22.58	9.58	9.93	9.62	9.30	8.88
Food away from home										
Mean	3,933	1,647	1,556	2,235	2,533	2,877	3,624	4,907	6,643	8,824
Share	5.1	5.1	4.4	4.8	5.0	4.8	5.0	5.4	5.6	5.0
SE	69.03	189.40	98.23	135.78	185.86	124.74	123.17	166.79	331.25	299.42
RSE	1.76	11.50	6.31	6.07	7.34	4.34	3.40	3.40	4.99	3.39

7. Nielsen only scan barcodes, so possibly only grocery no QSR or other FAFH

8. In model estimates

- pre-trend significantly reduced after being weighted (WLS)

Table 1: Model Estimates

Term	TWFE	TWFE Log	TWFE	TWFE Log	DDD	DDD Log
Post	1.9256*** (0.0988)	0.1343*** (0.0057)	2.4608*** (0.0878)	0.1713*** (0.0130)	2.0055*** (0.0979)	0.1398*** (0.0064)
State × Post	0.3908*** (0.1400)	0.0275** (0.0139)				
SNAP × Post			-1.0536*** (0.2135)	-0.0694*** (0.0085)		
State × SNAP × Post					-0.5983** (0.2394)	-0.0380*** (0.0127)
N	7,731,675	7,731,675	1,308,407	1,308,407	7,731,675	7,731,675
R ²	0.0026	0.0040	0.0035	0.0055	0.0026	0.0040
User FE	Yes	Yes	Yes	Yes	Yes	Yes
Period FE	No	No	No	No	No	No
SE Cluster	state	state	state	state	state	state

Table 7: Model Estimates

Term	TWFE	TWFE Log	TWFE	TWFE Log	DDD	DDD Log
Post	0.6516*** (0.0599)	0.0161*** (0.0030)	0.9034*** (0.1093)	0.0237*** (0.0078)	0.6867*** (0.0594)	0.0171*** (0.0029)
State × Post	0.1376 (0.1575)	0.0028 (0.0096)				
SNAP × Post			-0.9200** (0.4312)	-0.0390*** (0.0149)		
State × SNAP × Post					-0.7033 (0.4893)	-0.0325 (0.0207)
N	4,791,552	4,791,552	796,344	796,344	4,791,552	4,791,552
R ²	0.0002	0.0000	0.0003	0.0001	0.0002	0.0000
User FE	Yes	Yes	Yes	Yes	Yes	Yes
Period FE	No	No	No	No	No	No
SE Cluster	state	state	state	state	state	state

9. on average 17% household on SNAP each month; but when aggregate at annual level, 38% household ever used SNAP benefit

10. Some 0 food spending and total spending; 22 incidences (out of all month-hhd observations) with 0 spending in static panel; not sure why Numerator still include them in the static panel

11. Average Number of Months per User & 17.88 & 18.06 & 18.03 -> because around 30 percent households are replaced between cohort